Computer Networks

Summarized Notes: TCP/IP Protocol Suite & Signals

This document summarizes two topics: (1) Protocol Layering & the TCP/IP Protocol Suite, and (2) Signals (Analog vs Digital, Signal Characteristics). Each topic is presented with a definition, key concepts, and advantages/disadvantages where applicable.

Note: Notes for 'Lecture 1 – Computer Networks Introduction' could not be included because that file was not available for reading.

UNIT 1: Protocol Layering & TCP/IP Protocol Suite

1. Need for Layered Protocol

When communication between two parties is simple, it can be handled using a single, simple protocol operating at one layer. However, when the communication task becomes complex, it is divided among several layers, with a separate protocol assigned to each layer — this approach is called **protocol layering**.

Single Layer Protocol

A single layer protocol is used when communication is so simple that it can occur in just one layer. For example, when Maria talks to Ann directly, both know they should greet each other, keep the vocabulary suited to their friendship, avoid interrupting one another, keep the conversation a dialog (not a monologue), and exchange pleasantries before leaving.

Three-Layer Protocol (Example)

In a more complex scenario (e.g., Maria sending an encrypted letter to Ann by post), three layers are needed: Layer 3 (Listen/Talk – handles the plaintext message), Layer 2 (Encrypt/Decrypt – converts plaintext to ciphertext and back), and Layer 1 (Send/Receive mail – handles the physical mail through the postal system).

Principles of Protocol Layering

- **Bidirectional communication:** Each layer must be able to perform two opposite tasks, one in each direction (e.g., Layer 3 listens and talks; Layer 2 encrypts and decrypts; Layer 1 sends and receives).
- **Identical objects:** The object exchanged under a given layer must be identical at both communicating sites (e.g., plaintext under Layer 3, ciphertext under Layer 2, mail under Layer 1).

Logical Connection

Although data physically travels down through the layers on the sending side and up through the layers on the receiving side, each layer is imagined to have a direct, logical (virtual) connection with its peer layer on the other device.

Advantages of Protocol Layering

- Breaks down a complex communication task into smaller, manageable, independent modules.
- Each layer can be designed, developed, and modified without affecting other layers.
- Encourages standardization since each layer performs a well-defined function.
- Simplifies troubleshooting and testing, since problems can be isolated to a specific layer.

Disadvantages of Protocol Layering

- Adds overhead due to headers/trailers added at each layer.
- Can introduce processing delay as data passes through multiple layers.
- Some functions may be duplicated across layers, reducing efficiency.

2. TCP/IP Protocol Suite

TCP/IP (Transmission Control Protocol/Internet Protocol) is a protocol suite — a set of protocols organized into different layers — used in the Internet today. It is a **hierarchical** protocol, meaning each upper-layer protocol is supported by the services of one or more lower-layer protocols. It was originally defined with four layers built on top of the hardware, but is commonly presented today as a **five-layer model**: Application, Transport, Network (Internet), Data Link, and Physical.

Devices Involved in Communication

- **Hosts (source & destination):** involved in all five layers.
- **Link-layer switch:** involved only in the Data Link and Physical layers.
- **Router:** involved in the Network, Data Link, and Physical layers (no Transport or Application layer, since routing does not require them); a router deals with multiple Data Link/Physical layer pairs — one for each link it connects to.

Duty of Each Layer Group

- **Application, Transport, Network layers:** duty is **end-to-end** (host to host); domain is the whole internet.
- **Data Link, Physical layers:** duty is **hop-to-hop** (host to router, or router to router); domain is a single link.

Description of Each Layer

a) Physical Layer

Responsible for carrying individual bits (in a frame) across the physical link. Communication at this layer is still logical, since below it lies the transmission medium (cable or air), which carries electrical or optical signals — not the bits themselves. Various protocols transform bits into signals.

b) Data Link Layer

Responsible for moving a packet (datagram) through a single link (LAN/WAN, wired/wireless). TCP/IP does not define a specific data-link protocol — it supports all standard and proprietary link protocols. This layer encapsulates a datagram into a **frame**.

c) Network Layer

Responsible for host-to-host delivery and routing of packets, choosing the best path through the network via routers. The main protocol is **IP (Internet Protocol)**, which defines the datagram format and performs routing. IP is **connectionless** and provides no flow control, error control, or congestion control.

Helper protocols at the Network layer:

- **ICMP** – reports problems encountered while routing a packet.
- **IGMP** – helps IP with multicasting.
- **DHCP** – helps IP obtain a network-layer address for a host.
- **ARP** – helps IP find a router's link-layer (MAC) address when its network-layer address is known.

d) Transport Layer

Provides end-to-end (process-to-process) delivery of the message from the application layer of the source host to the application layer of the destination host. It encapsulates the message into a **segment** (TCP) or a **user datagram** (UDP).

e) Application Layer

Provides end-to-end, process-to-process communication between two application programs (a request is sent, and a response is received). Common application-layer protocols include:

- **HTTP** – accessing the World Wide Web.
- **SMTP** – sending electronic mail.
- **FTP** – transferring files between hosts.
- **TELNET / SSH** – remote access to a site.
- **SNMP** – network management at global and local levels.
- **DNS** – finding the network-layer address of a computer by name.
- **IGMP** – collecting group membership information.

3. Addressing

Any communication involving two parties needs a source address and a destination address. In TCP/IP, there are conceptually five pairs of addresses (one per layer), but only four are actually used because the Physical layer does not need addresses.

Layer	Packet Name	Address Type
Application	Message	Names
Transport	Segment / User Datagram	Port numbers
Network	Datagram	Logical (IP) addresses
Data Link	Frame	Link-layer (MAC) addresses
Physical	Bits	— (no address needed)

4. Multiplexing and Demultiplexing

Since TCP/IP uses several protocols at some layers, multiplexing and demultiplexing are required. **Multiplexing** means a protocol at one layer can encapsulate data from several next-higher-layer protocols (one at a time) — e.g., IP carries data from both TCP and UDP. **Demultiplexing** means a protocol can decapsulate and deliver data to the correct next-higher-layer protocol at the destination (e.g., IP delivers data up to either TCP or UDP based on header information).

5. Encapsulation and Decapsulation

Encapsulation and decapsulation is a core concept of protocol layering in the Internet: encapsulation happens at the source host, decapsulation happens at the destination host, and both encapsulation and decapsulation occur at each router along the path.

At the Source Host (Encapsulation)

- **Application layer:** data to be exchanged is called a **message**.
- **Transport layer:** adds a header (source/destination process IDs, flow/error/congestion-control info) to form a **segment** (TCP) or **user datagram** (UDP).
- **Network layer:** adds a header (source/destination host addresses, fragmentation info) to form a **datagram**.
- **Data Link layer:** adds a header with link-layer addresses to form a **frame**, which is passed to the physical layer for transmission.

At the Router (Decapsulation + Encapsulation)

- The Data Link layer decapsulates the datagram from the incoming frame and passes it to the Network layer.
- The Network layer inspects only the source/destination addresses and consults its forwarding table to find the next hop; the datagram's contents are not changed (unless fragmentation is required).
- The Data Link layer of the outgoing link re-encapsulates the datagram into a new frame for the next hop.

At the Destination Host (Decapsulation)

Each layer decapsulates the received packet, removes the payload, and passes it up to the next-higher layer until the original message reaches the application layer. Decapsulation at the host also involves error checking at each stage.

6. OSI Model vs TCP/IP Model

The OSI (Open Systems Interconnection) model has seven layers: Application, Presentation, Session, Transport, Network, Data Link, and Physical. The TCP/IP suite combines the top three OSI layers (Application, Presentation, Session) into a single **Application layer**, and keeps Transport, Network, Data Link, and Physical layers similar to OSI.

Why Session and Presentation Layers Are Missing in TCP/IP

- TCP/IP has more than one transport-layer protocol, and some session-layer functionality is already available within some transport-layer protocols.
- The application layer is not a single piece of software — many different applications can be built at this layer, and any needed presentation/session functionality can be included directly within that application's own software.

Why TCP/IP Became Dominant Over OSI

- OSI was completed only after TCP/IP was already fully implemented and widely deployed; switching would have cost too much time and money.
- Some OSI layers (particularly Presentation and Session) were never fully and completely defined or implemented in software.
- When implemented, OSI did not demonstrate high enough performance to convince the Internet community to abandon TCP/IP.

Advantages of TCP/IP Model

- Open, standardized, and vendor-independent — forms the backbone of the modern Internet.
- Fewer layers (compared to OSI) make it simpler to implement.
- Highly scalable and supports routing across large, heterogeneous networks.
- Well-tested and mature, with a huge base of real-world deployment.

Disadvantages of TCP/IP Model

- Does not clearly separate the concepts of services, interfaces, and protocols (unlike OSI).
- Not a general model — it is fitted specifically to the existing protocol suite, making it less suitable for describing other, non-TCP/IP protocol stacks.
- The Application layer combines functions (session, presentation, application) that OSI keeps separate, making its scope less precisely defined.

UNIT 2: Signals

Topics covered: Analog and Digital Signals, Characteristics of Analog Signals, Frequency Modulation (FM), Amplitude Modulation (AM), Phase Modulation (PM), and Quadrature Amplitude Modulation (QAM).

1. Definition of a Signal

A **signal** is a function of one or more independent variables that conveys some information. Signals are widely used in communication systems — e.g., telecommunication towers, telephones, radio transmission, human speech (listening/talking), and electronic messaging.

Classification by Time

- **Continuous-time signals, x(t):** defined at every instant of time (example: analog signals such as a smooth sine wave).
- **Discrete-time signals, x(n):** defined only at discrete instants of time (example: digital signals, shown as a series of separate sample points).

Example: ECG Signal

An electrocardiogram (ECG) signal is a real-world example of a continuous, time-varying signal, showing repeating waveform patterns (P, Q, R, S, T, U points) that represent the electrical activity of the heart.

Example: Voice-to-Digital Conversion

Human voice (an analog signal) can be converted into a digital signal using techniques such as **PCM (Pulse Code Modulation)** encoding, which samples the analog waveform and represents it as a sequence of numeric/binary values.

Example: Signals in an Electrical Circuit

In an RC circuit, the source voltage (v_s) and capacitor voltage (v_c) are both signals — they are patterns of variation over time, illustrating how a signal need not be a communication message but any quantity that varies with time.

2. Analog Signals vs Digital Signals

Signals can be classified as **analog** or **digital**. An analog signal can take an infinite number of values within a range, producing a smooth, continuous waveform. A digital signal can take only a limited (finite) number of values, producing a stepped waveform.

Advantages of Analog Signals

- Represent real-world continuous phenomena naturally and with high precision.
- Simpler circuitry historically required for generation and processing.
- No quantization error, since values vary continuously.

Disadvantages of Analog Signals

- Highly susceptible to noise and signal degradation over distance.
- Difficult to process, store, and reproduce without loss of quality.
- Harder to encrypt or secure compared to digital signals.

Advantages of Digital Signals

- More resistant to noise and interference; easier to regenerate accurately.
- Easier to store, compress, encrypt, and process using computers.
- Enables error detection and correction techniques.

Disadvantages of Digital Signals

- Requires sampling and quantization, which can introduce quantization error/loss of information.
- Needs more bandwidth in some cases compared to an equivalent analog transmission.
- Requires additional hardware/processing for analog-to-digital and digital-to-analog conversion.

3. Characteristics of Analog Signals

A simple analog signal, such as a **sine wave**, cannot be decomposed further, whereas a composite analog signal is made up of multiple sine waves. Analog (sine wave) signals are described using three main characteristics: amplitude, frequency (and period), and phase.

a) Amplitude

Amplitude refers to the height of the signal's value from its baseline at any given point — i.e., how large or small the signal's peak value is. Two signals with the same phase and frequency can still have different (high or low) peak amplitudes.

b) Frequency and Period

Frequency is the rate of change of a signal with respect to time. A change occurring over a short span of time means high frequency; a change occurring over a long span of time means low frequency. If a signal does not change at all, its frequency is zero; if it changes instantaneously, its frequency is considered infinite. **Period** is the amount of time (in seconds) needed to complete one cycle of the signal. Frequency and period are the inverse of each other:

$$f = 1 / T \quad \text{and} \quad T = 1 / f$$

Example problems from the lecture: (1) home power supply frequency is 60 Hz (or 50 Hz in Europe) — find the period; (2) express a period of 100 ms in microseconds; (3) given a period of 100 ms, find the frequency in kilohertz.

c) Phase

Phase describes the position of the waveform relative to time 0 — i.e., how the wave is shifted in relation to the start of the time axis.

- A sine wave with a **phase of 0°** starts at time 0 with zero amplitude, and the amplitude is increasing; it is not shifted.
- A sine wave with a **phase of 90°** starts at time 0 with peak amplitude, and the amplitude is decreasing; the signal is shifted to the left by a quarter cycle and does not really exist before time 0.
- A sine wave with a **phase of 180°** starts at time 0 with zero amplitude, and the amplitude is decreasing; the signal is shifted to the left by half a cycle and does not really exist before time 0.

d) Wavelength

Wavelength is the distance a simple signal travels in one period through a transmission medium. It relates the period of the signal to the propagation speed of the medium, and describes how much physical distance one complete cycle of the wave occupies as it propagates.

4. The Taxonomy of Signals

Signals can be classified along two dimensions — **Time** (continuous or discrete) and **Amplitude** (continuous or discrete) — giving four categories:

Time \ Amplitude	Continuous Amplitude	Discrete Amplitude
Continuous Time	Analog signals	Continuous-time signals
Discrete Time	Discrete-time signals	Digital signals

5. Types of Modulation

Modulation is the process of varying a property of a carrier (analog) signal in order to encode information for transmission. The lecture syllabus identifies four main types:

- **Frequency Modulation (FM):** information is encoded by varying the **frequency** of the carrier signal in proportion to the message signal.
- **Amplitude Modulation (AM):** information is encoded by varying the **amplitude** of the carrier signal in proportion to the message signal.
- **Phase Modulation (PM):** information is encoded by varying the **phase** of the carrier signal in proportion to the message signal.
- **Quadrature Amplitude Modulation (QAM):** combines amplitude and phase modulation, transmitting two signals simultaneously by modulating the amplitudes of two carrier waves that are 90° (quadrature) out of phase with each other — used to increase data-carrying capacity.

General Advantages of Modulation

- Allows signals to be transmitted efficiently over long distances using appropriately sized antennas.
- Enables multiple signals to share the same medium via multiplexing (different carrier frequencies).
- Can improve noise immunity (particularly FM/PM/QAM) compared to transmitting baseband signals directly.

General Disadvantages of Modulation

- Requires additional hardware (modulator/demodulator) at transmitter and receiver, adding cost and complexity.
- Some schemes (e.g., AM) are more susceptible to noise and amplitude-based interference.
- Higher-order schemes (e.g., QAM) are more sensitive to channel impairments and require more precise synchronization.